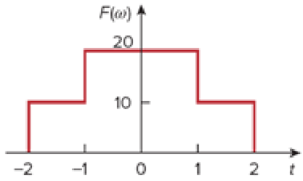
Problem

Determine the signal $f(t)$ whose Fourier transform is shown in Fig. 18.38. (*Hint:* Use the duality property.)

Figure 18.38



Step-by-step solution

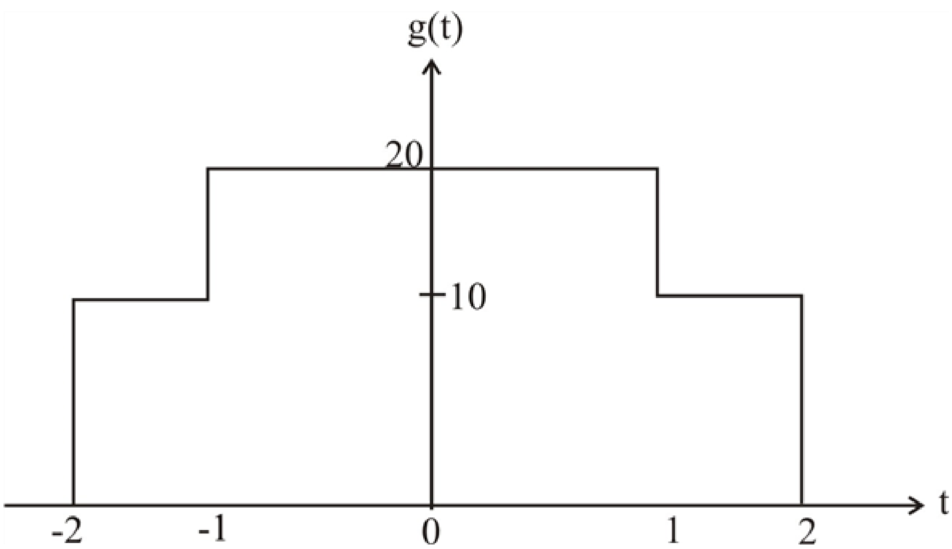
Step 1 of 4

Firsdt, we find $G(\omega)$ for $g(t)$ shown below,

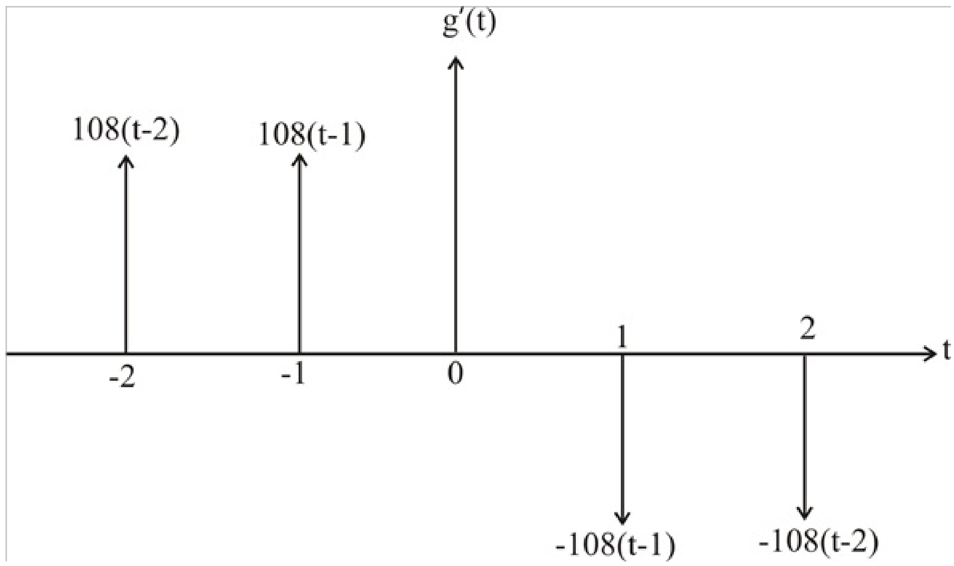
$$g(t) = 10\left[u(t+2) - u(t-2)\right] + 10\left[u(t+1) - u(t-1)\right]$$

$$g'(t) = 10\left[s(t+2) - s(t-2)\right] + 10\left[s(t+1) - s(t-1)\right]$$

Step 2 of 4



Step 3 of 4



Step 4 of 4

$$j\omega G(\omega) = 10\left(e^{j\omega z} - e^{-j\omega z}\right) + 10\left(e^{j\omega} - e^{-j\omega}\right)$$
$$= 20j\sin 2\omega + 20j\sin\omega$$
$$G(\omega) = \frac{20\sin\omega}{\omega} + \frac{20\sin 2\omega}{\omega} = 40\sin C(2\omega) + 20\sin C(\omega)$$

Note that

$$G(\omega) = G(-\omega)$$
$$F(\omega) = 2\pi G(-\omega)$$
$$f(t) = \frac{1}{2\pi}G(t)$$
$$= \frac{20}{\pi}\sin C(2t) + \frac{10}{\pi}\sin Ct$$